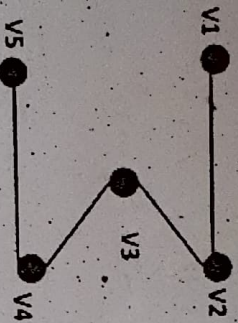


UNIT IV

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17. Obtain the linked list representation, the Adjacency matrix and the incidence matrix of the following graph: (11)



Or

18. Explain BPS and DFS graph traversal with an example. (11)

UNIT V

19. Explain the following in the design of hashing techniques
- (a) Division method (5)
 - (b) Folding method. (6)

Or

20. Explain the following in the design of file organization
- (a) Sequential organization (5)
 - (b) Indexing techniques (6)

B.Tech. DEGREE EXAMINATION, NOVEMBER 2017.

Third Semester

Computer Science of Engineering

DATA STRUCTURES

(2013 – 2014 onwards)

Time : Three hours

Maximum : 75 marks

PART A – (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. List out any two examples of linear and non linear data structures.
2. What are all the operations that can be performed on an array?
3. Find the value of the expression $752 + 415 - / -$ in postfix notation.
4. What are all the various structures of Queue?
5. Draw a binary tree with 9 nodes.

6. List out any two advantages of threaded binary tree.
7. Draw a complete graph with 4 vertices.
8. In what situation topological ordering is not possible in a graph. Justify?
9. Define symbol table.
10. List out any four external storage devices.

PART B — (5 × 11 = 55 marks)

Answer ALL the Units, choosing One questions from each Unit.

All questions carry equal marks.

UNIT I

11. (a) Illustrate the recursive implementation of Binary Search algorithm. (5)
- (b) Explain Quick sort algorithm and trace the algorithm with the following list as sample 25, 5, 37, 1, 61, 11, 59, 15, 48, 19. (6)

Or

12. (a) Illustrate the algorithm Fibonacci Search (5).

- (b) Explain two way Merge sort algorithm and trace the algorithm with the following list as sample 26, 5, 77, 1, 61, 11, 59, 15, 49, 19. (6)

UNIT II

13. (a) Specify the algorithms for the basic operations required to manipulate a Stacks. (5)
- (b) With neat diagram explain the three cases of how to delete a node from a single linked list. (6)

Or

14. (a) Specify the algorithms for queue insertion (ENQUEUE) and deletion (DEQUEUE) operation. (6)
- (b) Write an algorithm to create a linked list representation of a polynomial having three variables. (6)

UNIT III

15. For the arithmetic expression $((A * (B - C)) / ((D - E) * (F + G - H)))$ construct the binary tree and apply the three tree traversal algorithms and obtain the results of traversals. (11)

Or

16. Explain in detail the concept of B tree and construct a B tree of order 3 with the following set of data and insert them in the order as they appear. 30, 60, 50, 40, 10, 70, 20, 80, 90. (11)

8. (a) Explain insertion sort. Discuss and how the passes of insertion sort for the following input 9,8,7,5,4,3,2,1. (7)

- (b) Distinguish the following sorting algorithms: Bubble sort and Heap sort. (4)

UNIT V

9. What are the operations on a B Tree? Explain. (11)

Or

10. State and explain the operations on Trie Tree. (11)

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2017.

Third Semester

Computer Science and Engineering

DATA STRUCTURES

(2009-12 Batches)

Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL questions.

1. What do you mean by data structure? Give example.
2. State the two phases of performance evaluation.
3. What are the difference between stack and Queue?
4. What do you mean by Garbage collection?
5. List the applications of Tree.
6. Define Graph.
7. State the drawbacks of inverted tables.
8. List the factors to be considered while choosing a sorting algorithm.
9. Compare B Tree and B+ Tree.
10. What do you mean by Trie Tree indexing?

SECTION B — (5 × 11 = 55 marks)

Answer ALL units choosing any ONE question from each Unit.

UNIT I

11. (a) What are the criteria used to analyze program? Explain. (6)
- (b) Discuss in detail about linear search with example. (5)

Or

12. (a) What is searching? Explain Binary search with example. (6)
- (b) Define array. Explain about multidimensional array in detail. (5)

UNIT II

13. (a) Describe about Dynamic Storage Management. (7)
- (b) Draw a structure of a typical language implementation. (4)

Or

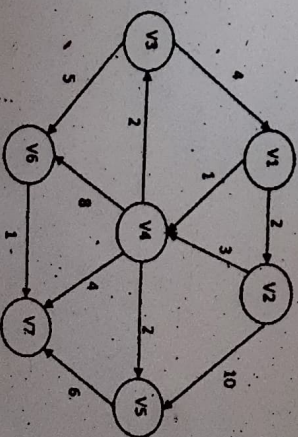
14. (a) Compare Linked Stack and Linked Queue. (4)
- (b) List the different types of linked list. Explain the concept of doubly linked list. (7)

UNIT III

15. (a) Explain the Traversals of binary tree with examples. (6)
- (b) What do you mean sort? Explain the topological sort procedure with an example. (5)

Or

16. Define Dijkstra's algorithm and find the shortest path from V_1 to all other vertices in a graph. (11)



UNIT IV

17. (a) Sort the following: 35, 45, 25, 11, 6, 85, 17, 35, 11 using shell sort. (6)
- (b) Explain the Quick sort algorithm with an example. (5)

Or